miniRT

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1 Linear algebra

1.1 Vector

A vector is a mathematical object described by it's components, it has 1 component for each dimension, so a 3D vector has 3 components.
It has the following operations for 2 vectors, $\vec{v}$, $\vec{w} \in \mathbb{R}^3$ and a scalar $\lambda \in \mathbb{R}$.

Addition	$\vec{v} + \vec{w}$	$=$	$(v_x + w_x, v_y + w_y, v_z + w_z)$
Substraction	$\vec{v} - \vec{w}$	$=$	$(v_x - w_x, v_y - w_y, v_z - w_z)$
Scale	$\lambda\vec{v}$	$=$	$(v_x \cdot \lambda, v_y \cdot \lambda, v_z \cdot \lambda)$
Dot product	$\vec{v} \cdot \vec{w}$	$=$	$v_x w_x + v_y w_y + v_z w_z$
Inner-product	$\langle \vec{v}, \vec{w} \rangle$	$=$	$\vec{v} \cdot \vec{w}$
Cross product	$\vec{v} \times \vec{w}$	$=$	$(v_y w_z - v_z w_y, v_z w_x - v_x w_z, v_x w_y - v_y w_x)$
Magnitude	$\|\|\vec{v}\|\|$	$=$	$\sqrt{v_x^2 + v_y^2 + v_z^2}$
Normalization	$\hat{v}$	$=$	$\vec{v} / \|\|\vec{v}\|\|$

1.2 Matrix

Respectively yaw, pitch and roll matrices :

$$R_z(\psi) = \begin{bmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad R_y(\theta) = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix} \qquad R_x(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi & \cos\phi \end{bmatrix}$$

2 Vector space objects

2.1 Ray

For a point $o \in \mathbb{R}^3$, the origin of the ray, a vector $\hat{d} \in \mathbb{R}^3$, the normalalized direction vector of the ray, and a scalar for the direction vector, $t \in \mathbb{R}$. We define the normal equation of a line as :

$$R(t) = o + t\hat{d}$$

2.2 Plane

For a point $p \in \mathbb{R}^3$, the origin of the plane, a vector $\hat{v} \in \mathbb{R}^3$, the normalized orthogonal vector of the plane and a arbitrary point $x \in \mathbb{R}^3$, it is a point of the plane if :

$$(x - p) \cdot \hat{v} = 0$$

2.3 Sphere

For a point $p \in \mathbb{R}^3$ the origin of the sphere, a scalar $r \in \mathbb{R}$, the radius of the sphere and an arbitrary point $x \in \mathbb{R}^3$., it is a point of the sphere if :

$$||x - p||^2 = r^2$$

2.4 Circle

For a point $p \in \mathbb{R}^3$, the origin of the circle, a vector $\hat{v} \in \mathbb{R}^3$, the normalized orthogonal vector of the circle and a arbitrary point $x \in \mathbb{R}^3$, it is a point of the circle if $||x - o|| \leq r$:

$$(x - p) \cdot \hat{v} = 0$$

Limit the surface of the plane to a radius of r :

$$||(x - o)|| \leq r$$

2.5 Cylinder

For a point $p \in \mathbb{R}^3$ the origin of the sphere, a a vector $\hat{v} \in \mathbb{R}^3$ the normalized direction vector, two scalars r and $h \in \mathbb{R}$, respectively the radius and the height of the cylinder. We define the normal equation of an infinite cylinder as :

$$||(x - p) - ((x - p) \cdot \hat{v})\hat{v}||^2 = r^2$$

Limit the height with :

$$-\frac{h}{2} \leq (r(t) - c) \cdot \hat{d} \leq \frac{h}{2}$$

And then use the previous formula for a circle to set the top and bottom cap of the cylinder.

2.6 Cone

For a point $p \in \mathbb{R}^3$ the origin of the cone, a a vector $\hat{v} \in \mathbb{R}^3$ the normalized direction vector. We define equation of an infinite cone as :

$$((x - p) \cdot \hat{v})^2 = \cos^2(\theta)||x - c||^2$$

3 Raytracer

3.1 Camera

Given informations :

- $\mathbf{C_{pos}}$: 3D Position as a point.
- $\mathbf{C_{dir}}$: 3D Direction as a normalized vector.
- fov : Vertical field of view in degrees, ranging from 0° to 180°.

Deduced informations :

- $\mathbf{C_{fov}}$: Vertical field of view in radians, ranging from 0 to π radians. We can define it with the following fomula :
$$\mathrm{C_{fov}} = \mathrm{fov} \cdot \frac{\pi}{180°}$$
- $\mathbf{C_{flen}}$: Focal length that can be calculated as follows :
$$\mathrm{C_{flen}} = \tan\left(\frac{\mathrm{C_{fov}}}{2}\right)$$
- $\mathbf{S_{ratio}}$: Pixel screen ratio defined by $\frac{\mathrm{Width}}{\mathrm{Height}}$

3.1.1 Camera rays

Calculate a ray passing from the camera's origin and a pixel of the screen :

$$\mathbf{S_{dir}} = \left[\left(2\left(\frac{x+0.5}{\text{Width}}\right)-1\right)\text{S}_{\text{ratio}}\text{C}_{\text{flen}}, \left(1-2\left(\frac{y+0.5}{\text{Height}}\right)-1\right)\text{C}_{\text{flen}}, 1\right]$$

Calculate camera's right and up vector :

$$\mathbf{C_{right}} = (0,1,0) \times \text{C}_{\text{dir}} \text{ and } \mathbf{C_{up}} = \text{C}_{\text{dir}} \times \text{C}_{\text{right}}$$

Get the ray shot through the given screen coordinate in space where $\mathbf{R_{pos}} = \text{C}_{\text{pos}}$ and

$$\mathbf{R_{dir}} = \begin{bmatrix} \hat{\text{C}}_{\text{right}_x} & \hat{\text{C}}_{\text{up}_x} & \hat{\text{C}}_{\text{dir}_x} \\ \hat{\text{C}}_{\text{right}_y} & \hat{\text{C}}_{\text{up}_y} & \hat{\text{C}}_{\text{dir}_y} \\ \hat{\text{C}}_{\text{right}_z} & \hat{\text{C}}_{\text{up}_z} & \hat{\text{C}}_{\text{dir}_z} \end{bmatrix} \begin{bmatrix} \text{S}_{\text{dir}_x} \\ \text{S}_{\text{dir}_y} \\ 1 \end{bmatrix}$$

3.2 Ray intersections

For the previously defined variables and object equations we can define the following intersection equations.
For second degree equations, you can find the distance t of the nearest positive root with :

$$t = \min\left\{\frac{-b-\sqrt{b^2-4ac}}{2a}, \frac{-b+\sqrt{b^2-4ac}}{2a}\right\}_{t>0}$$

3.2.1 Plane

$$t = \frac{(p-o)\cdot\hat{v}}{\hat{d}\cdot\hat{v}}$$

t is the distance between the plane and the ray origin followed by it's direction vector if it is a finite result, and if $t < 0$ the plane lays behind the ray origin.

3.2.2 Circle

Same as a plane but where you have to check wether or not the hitpoint - circle origin vector is less or equal to the radius of the circle.

3.2.3 Sphere

If you substitute the equation of a ray into the one of the sphere you obtain a second degree equation :

$$||(o+t\hat{d})-p||^2 = r^2 \iff (\hat{d}\cdot\hat{d})t^2 + 2((o-p)\cdot\hat{d})t + ((o-p)\cdot(o-p)-r^2) = 0$$

$$\begin{aligned} a &= \hat{d}\cdot\hat{d} \\ b &= (o-p)\cdot\hat{d} \\ c &= (o-p)\cdot(o-p)-r^2 \end{aligned}$$

3.2.4 Cylinder

If you substitute the equation of a ray into the one of the cylinder you obtain a second degree equation :

$$||((o+t\hat{d})-p)-(((o+t\hat{d})-p)\cdot\hat{v})\hat{v}||^2 = r^2 \iff (\hat{d}\cdot\hat{d})t^2+2((o-p)\cdot\hat{d})t+((o-p)\cdot(o-p)-r^2) = 0$$

$$\begin{aligned} a &= \hat{d}\cdot\hat{d} \\ b &= (o-p)\cdot\hat{d} \\ c &= (o-p)\cdot(o-p)-r^2 \end{aligned}$$

3.2.5 Cone

If you substitute the equation of a ray into the one of the cone you obtain a second degree equation :

$$((o+t\hat{d}-p)\cdot\hat{v})^2 = \cos^2(\theta)||o+t\hat{d}-c||^2 \iff$$

$$((\hat{d}\cdot\hat{v})^2-\cos^2(\theta))t^2+(2((\hat{d}\cdot\hat{v})((o-p)\cdot\hat{v})-\cos^2(\theta)(\hat{d}\cdot(o-p))))t+((o-p)\cdot\hat{v})^2-\cos^2(\theta)||(o-p)||^2 = 0$$

$$\begin{aligned} a &= (\hat{d}\cdot\hat{v})^2-\cos^2(\theta) \\ b &= 2((\hat{d}\cdot\hat{v})((o-p)\cdot\hat{v})-\cos^2(\theta)(\hat{d}\cdot(o-p))) \\ c &= ((o-p)\cdot\hat{v})^2-\cos^2(\theta)||(o-p)||^2 \end{aligned}$$